

```
In [151... import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from scipy.optimize import curve_fit
```

```
In [153... data = pd.read_csv("data/data.csv")
```

```
In [155... data.head()
```

```
Out[155...
      VideoAnalysis:  VideoAnalysis:  VideoAnalysis:  VideoAnalysis:  VideoAnalysis:
      Time (s)      X (m)      Y (m)      X Velocity      Y Velocity
      (m/s)      (m/s)
0      0.595      2.315135      7.353513      12.297397      26.324326
1      0.629      2.751815      8.270541      11.751259      25.351348
2      0.662      3.101159      9.100233      11.452038      23.913709
3      0.695      3.494172      9.842590      11.506013      22.579152
4      0.729      3.887184      10.584946      11.161415      21.924713
```

Since for calculation purpose it is easier to start at time = 0
we are going to subtract t initial from all the time values

```
In [158... data['VideoAnalysis: Time (s)'] -= data['VideoAnalysis: Time (s)'][0]
```

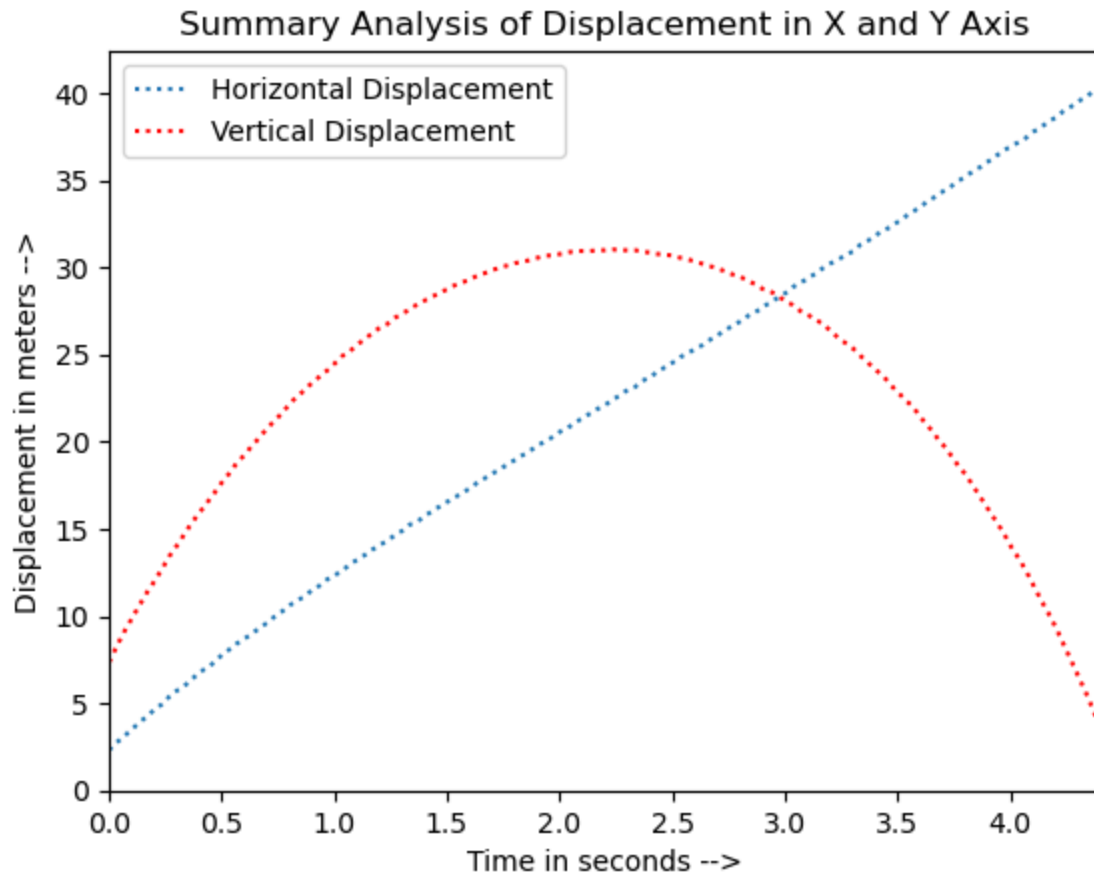
```
In [159... data.head()
```

```
Out[159...
      VideoAnalysis:  VideoAnalysis:  VideoAnalysis:  VideoAnalysis:  VideoAnalysis:
      Time (s)      X (m)      Y (m)      X Velocity      Y Velocity
      (m/s)      (m/s)
0      0.000      2.315135      7.353513      12.297397      26.324326
1      0.034      2.751815      8.270541      11.751259      25.351348
2      0.067      3.101159      9.100233      11.452038      23.913709
3      0.100      3.494172      9.842590      11.506013      22.579152
4      0.134      3.887184      10.584946      11.161415      21.924713
```

```
In [161... fig, ax = plt.subplots()
ax.set_title("Summary Analysis of Displacement in X and Y Axis")
ax.set_xlabel("Time in seconds -->")
ax.set_ylabel("Displacement in meters -->")
ax.plot(data.iloc[:, 0], data.iloc[:, 1], label = "Horizontal Displacement", l
ax.plot(data.iloc[:, 0], data.iloc[:, 2], label = "Vertical Displacement", lin

# Minimum and Maximum element of time set as the x axis limit
```

```
ax.set_xlim(min(data['VideoAnalysis: Time (s)']), max(data['VideoAnalysis: Time (s)']))  
ax.set_ylim(bottom = 0)  
ax.legend()  
pass
```



In [163... data

Out[163...

	VideoAnalysis: Time (s)	VideoAnalysis: X (m)	VideoAnalysis: Y (m)	VideoAnalysis: X Velocity (m/s)	VideoAnalysis: Y Velocity (m/s)
0	0.000	2.315135	7.353513	12.297397	26.324326
1	0.034	2.751815	8.270541	11.751259	25.351348
2	0.067	3.101159	9.100233	11.452038	23.913709
3	0.100	3.494172	9.842590	11.506013	22.579152
4	0.134	3.887184	10.584946	11.161415	21.924713
...	...	...	...	...	...
128	4.271	39.214610	7.484517	8.624977	-28.012708
129	4.304	39.520287	6.523820	8.625520	-28.737792
130	4.338	39.782295	5.563124	8.935645	-29.505061
131	4.371	40.087971	4.558759	10.082347	-30.571791
132	4.404	40.480983	3.510727	11.084619	-31.266742

133 rows × 5 columns

Measuring the cone in the lab we found out the height to be 1.22 meters and measuring the cone in Logger Pro we got 1.31 meters as the height

The uncertainty is $(1.31 - 1.22)$ meters = 0.5 meters

The uncertainty of the ball movemet is approximatex 0.5 meters because on each frame change the ball tends to move around 0.5 meters.

Adding these two we get $(0.5 + 0.5)$ meters = 1 meters

total uncertainty in y = 1 meters

For the uncertainty in the x axis we can take the time difference between two measurments as an uncertainty and the logger

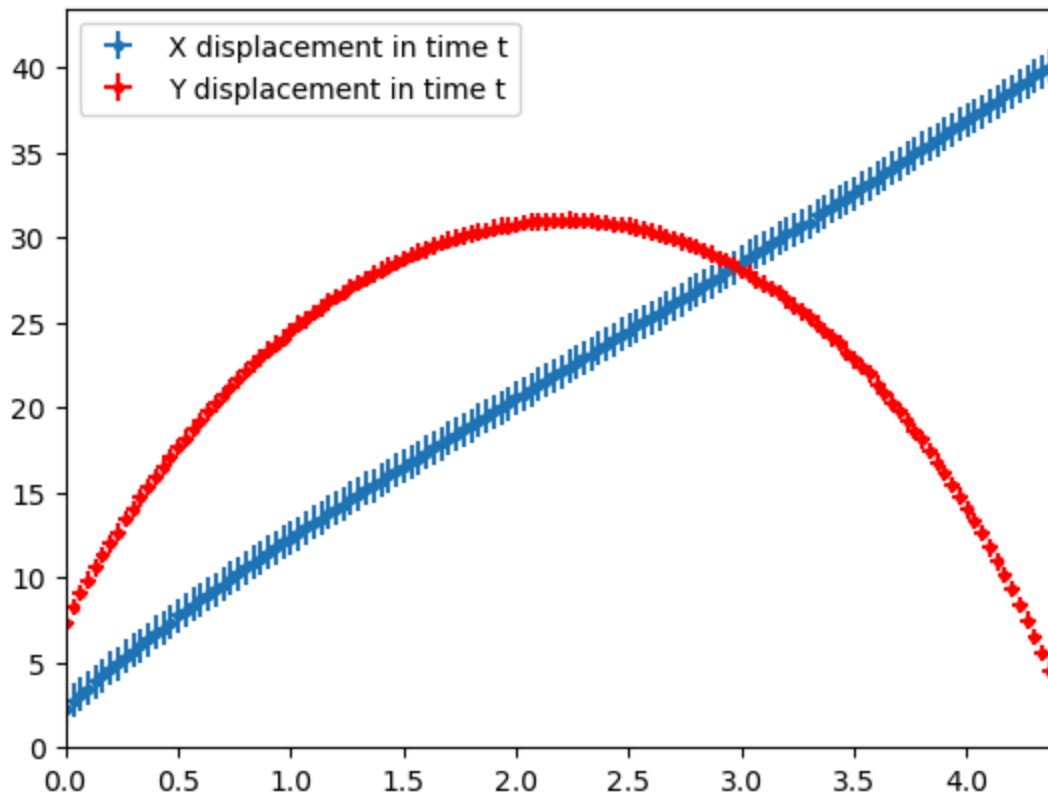
which is 0.03 seconds

In [377...

```
y_uncertainty = 0.5 + 0.5
x_uncertainty = 0.3
```

```
In [379... fig, ax = plt.subplots()
ax.set_title("Horizontal and Vertical Displacement with associated uncertainty
ax.errorbar(data.iloc[:, 0], data.iloc[:, 1], label = "X displacement in time
ax.errorbar(data.iloc[:, 0], data.iloc[:, 2], label = "Y displacement in time
ax.legend()
ax.set_xlim(min(data.iloc[:, 0]), max(data.iloc[:, 0]))
ax.set_ylim(bottom = 0)
ax.legend()
pass
```

Horizontal and Vertical Displacement with associated uncertainty



Dominant Source of uncertainty:

The dominant source of uncertainty is our eyes getting not tracking the exact ball movement

```
In [382... # model used to fit the velocity along y axis
linear_model = lambda x, A, B: x * A + B
quad_model = lambda x, A, B, C: A * x**2 + B * x + C
```

```
In [384... linear_parameters, cov = curve_fit(linear_model, data["VideoAnalysis: Time (s)
quadratic_parameters, cov = curve_fit(quad_model, data["VideoAnalysis: Time (s)
```

```
In [386... print("Linear parameters:", linear_parameters)
print("Quadratic parameters:", quadratic_parameters)
v_x = linear_parameters[0]
x_intercept = linear_parameters[1]
```

```
v_y = quadratic_parameters[1]
y_intercept = quadratic_parameters[2]
```

Linear parameters: [8.34050841 3.62377926]

Quadratic parameters: [-5.11719818 22.07262371 7.56490506]

Physics Connection to Parameters

- The linear fit shows a constant velocity of **v_x** m/s and an initial position of **x_intercept** meters.
- The quadratic fit suggests an initial vertical velocity of **v_y** m/s and an initial height of **y_intercept** meter

slope of horizontal displacement = 8.3 (m/s) +/- 1 (m/s)

X Intercecpt = 3.6 (m/s) +/- 1 (m/s)

slope of vertical displacement = 22.0 (m/s) +/- 1 (m/s)

Y Interept = 7.6 (m/s) +/- 1 (m/s)

The angle of the launch is give by

$$\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right)$$

```
In [391... theta = np.arctan(v_y / v_x)
theta = np.degrees(theta)
print("Angle =", theta)
```

Angle = 69.30009326532154

Since the dominant source of the uncertainty is relatively very large compared to other uncertainties we would only consider that in our estimations

Angle of launch = 69.3 (degree) +/- 1 (degree)

We cana get the range by

$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

```
In [404... v_inital = np.sqrt(v_x ** 2 + v_y**2)
gravity = 9.81
```

```
range = ((v_initial)**2 * np.sin(np.radians(2 * theta)) / gravity) + x_intercept
print(f"Range = {range:.2g}")
```

Range = 41

The calculated range is

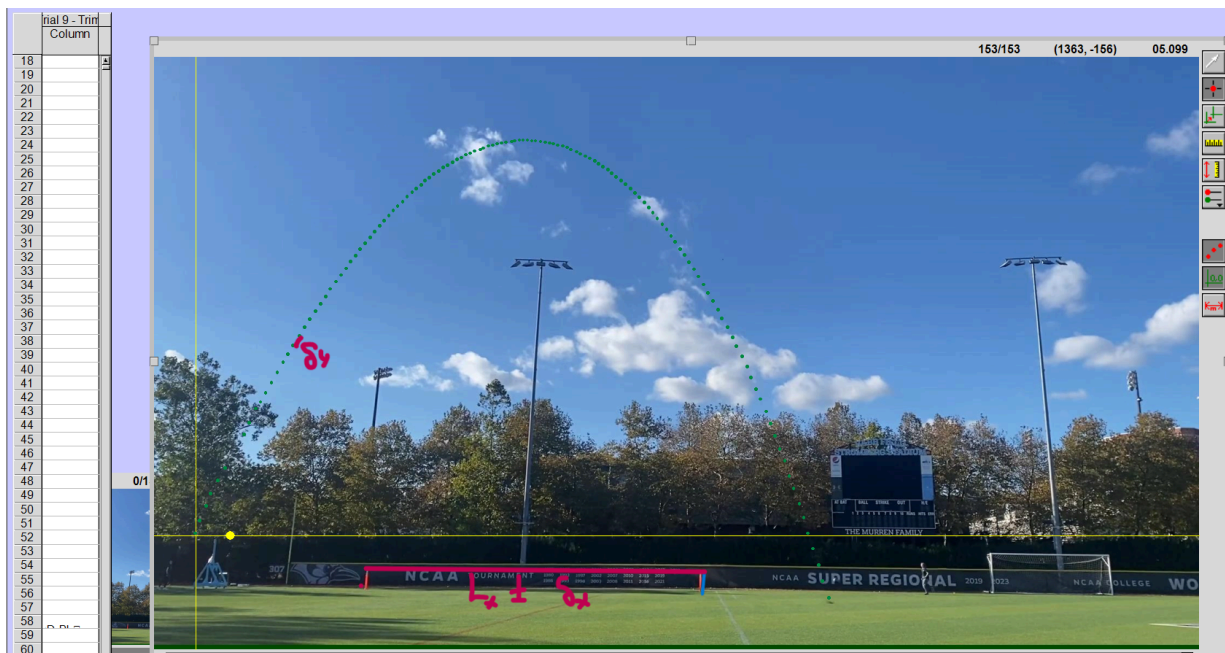
$$\text{Range} = 41 \pm 1 \text{ meters}$$

Comparison between logger pro results and our results in python:

In logger pro by fitting the curve fit we got the slope of the horizontal displacement to be 8.4 and the intercept to be -1.4 meter. For the Y motion we got the slope of the vertical displacement to be 22.4 meters/second and the intercept to be -7.4 meters. For the range the distance between where the ball was launched and where it landed was 39 meters.

The results seem to be close to our measurements.

```
In [2]: from IPython.display import Image, display
display(Image('images/Screenshot 2024-10-14 191626.png'))
```



```
In [ ]:
```